

Power & TIE plots

2026-03-12

```
library(ggplot2)
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

options(scipen = 999)

# Load data ----
setwd("/Users/remus/Documents/GitHub/latent_interaction/post_sim_review/binary_AERA")
load(file = "agg_results.rda")

# Get unique combinations of rsq_prod and n_items
combinations <- agg_results %>%
  distinct(rsq_prod, n_items)

# Loop over each combination
for (i in 1:nrow(combinations)) {

  plot_data <- agg_results %>%
    filter(rsq_prod == combinations$rsq_prod[i], n_items == combinations$n_items[i],
           group_prob %in% c("50/50", "80/20"))

  # Skip if no data
  if (nrow(plot_data) == 0) next

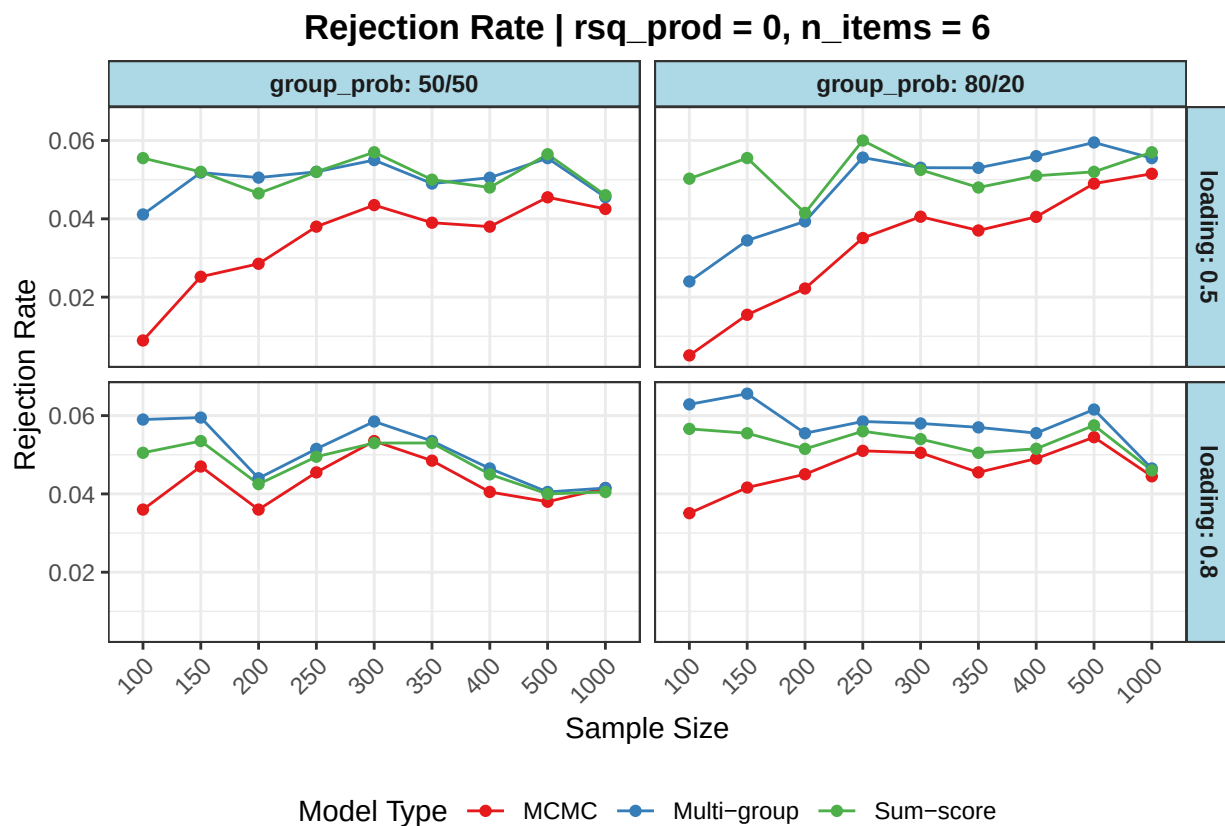
  plot_data <- plot_data %>%
    mutate(
      group_prob = factor(group_prob, levels = c("50/50", "80/20")),
      loading     = factor(loading),
      model_type  = factor(model_type),
      sample_size = factor(sample_size))
}
```

```

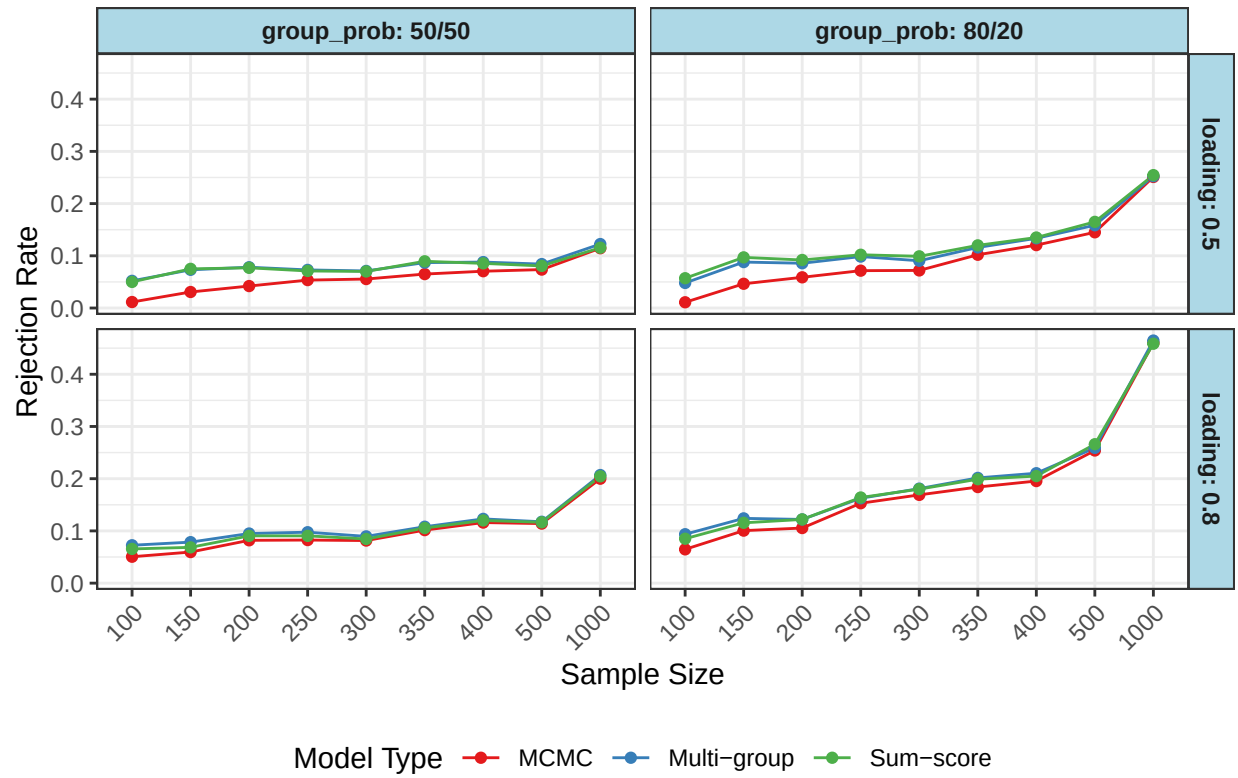
p <- ggplot(plot_data, aes(x = sample_size, y = sig,
                           color = model_type,
                           group = model_type)) +
  geom_line() + geom_point() + facet_grid(rows = vars(loading),
                                           cols = vars(group_prob),
                                           labeller = label_both) +
  scale_color_brewer(palette = "Set1", name = "Model Type") +
  labs(
    title = paste0("Rejection Rate | rsq_prod = ", combinations$rsq_prod[i],
                  ", n_items = ", combinations$n_items[i]),
    x = "Sample Size", y = "Rejection Rate") +
  theme_bw() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    legend.position = "bottom",
    axis.text.x = element_text(angle = 45, hjust = 1),
    strip.background = element_rect(fill = "lightblue"),
    strip.text = element_text(face = "bold"))

print(p)
}

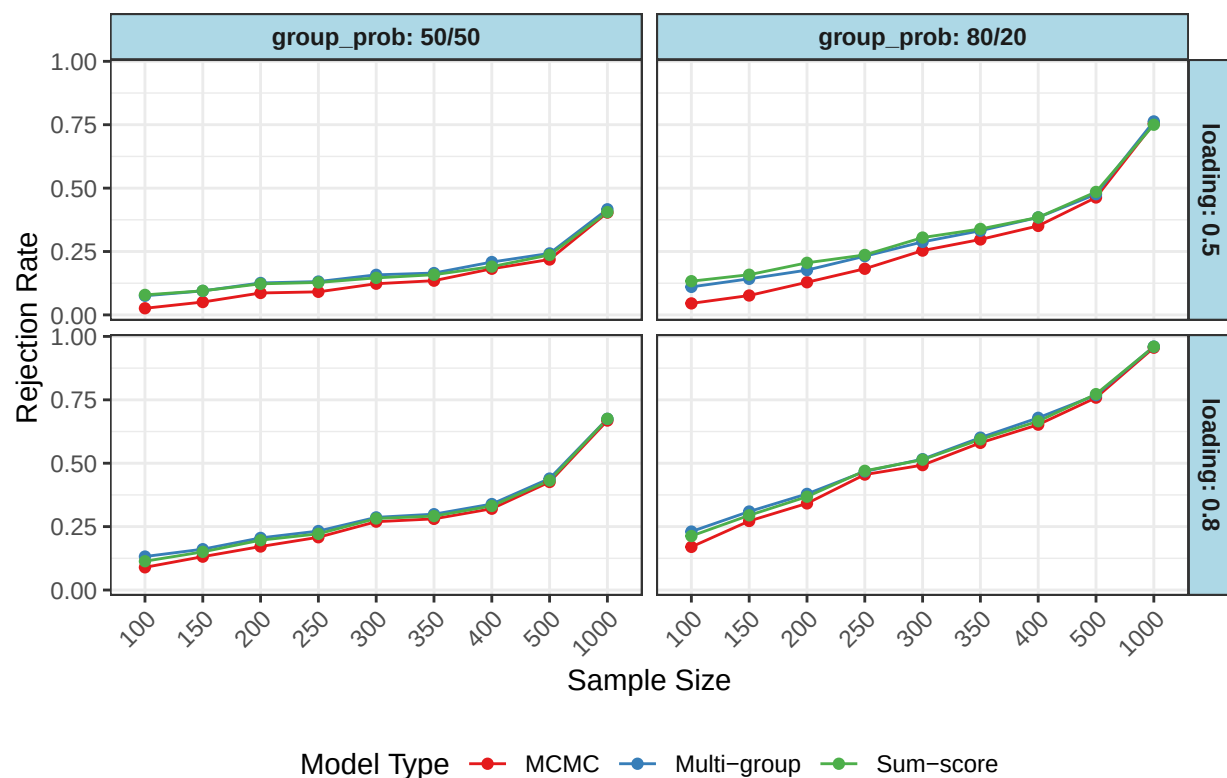
```



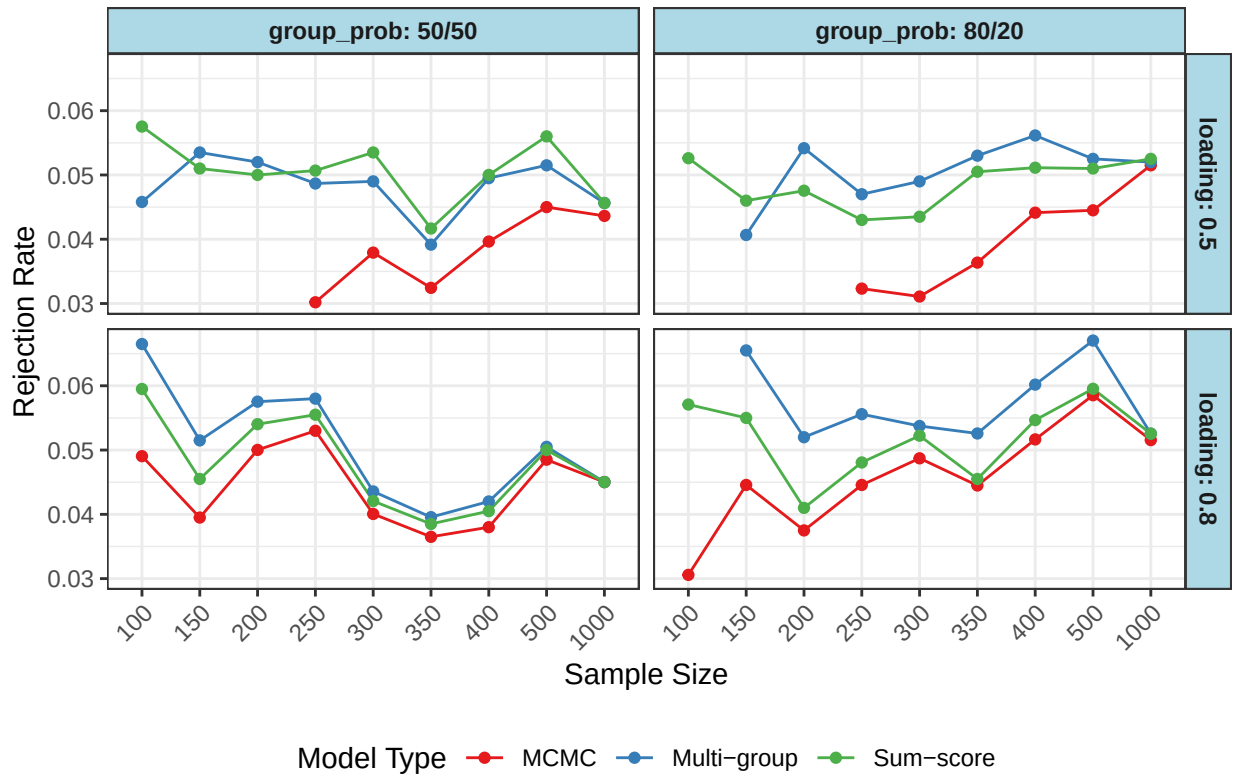
Rejection Rate | $rsq_prod = 0.03$, $n_items = 6$



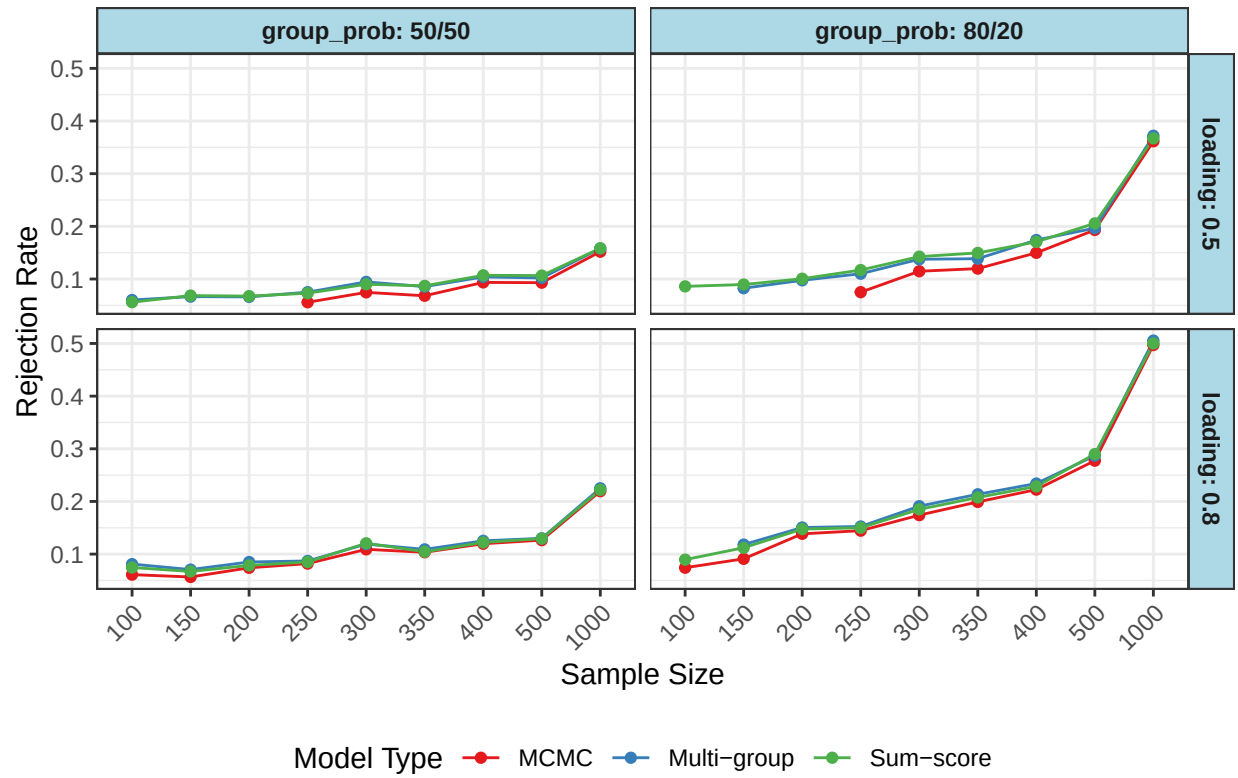
Rejection Rate | rsq_prod = 0.07, n_items = 6



Rejection Rate | rsq_prod = 0, n_items = 12



Rejection Rate | rsq_prod = 0.03, n_items = 12



Rejection Rate | $rsq_prod = 0.07$, $n_items = 12$

